Multimedia Signal Processing - 2nd module

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July, 6 2011

1 Question - 10 pts

Consider a source X that emits the following sequence of symbols:

$$\{1, 2, 6, 4, 10, 11, 10\}$$

1. Consider a quantizer Q_1, characterized by the following intervals and reconstruction levels
   - $I_1 = [-\infty, -1)$, $y_1 = -2$
   - $I_2 = [-1, 0)$, $y_2 = -0.5$
   - $I_3 = [0, +1)$, $y_3 = +0.5$
   - $I_4 = [+1, +\infty)$, $y_4 = +2$

   Plot the quantization function $y = Q_1(x)$. Encode the sequence with DPCM, quantizing the prediction residuals with Q_1 (assume that the first symbol is lossless coded). Compute the MSE distortion.
2. Design a Lloyd-Max quantizer for the source X. Initialize the algorithm with the following intervals
   - $I_1 = [-\infty, +3)$
   - $I_2 = [+3, +8)$
   - $I_3 = [+8, +10.5)$
   - $I_4 = [+10.5, +\infty)$

   Compute the MSE distortion upon convergence.

2 Question - 5 pts

Illustrate the block diagram of a perceptual audio codec (both encoder and decoder).

Describe how the perceptual model exploits frequency masking.

3 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Let X denote a continuous source, and $D(R)$ the corresponding rate-distortion function, and σ_x^2 its variance.
   - [T] [F] $H(X) < \infty$
   - [T] [F] $D(0) = \sigma_x^2$
   - [T] [F] $R(D) \leq \frac{1}{2} \log_2 \frac{\sigma_x^2}{D}$
2. Consider a speech coding system
   - [T] [F] Perceptual weighting might be used to take into account the masking effect of formants.
   - [T] [F] Direct quantization of prediction coefficients might lead to instability at the decoder.
   - [T] [F] In open-loop vocoders, the output signal approximates the samples of the input speech waveform.
3. Consider a hybrid DPCM/DCT video coding system
   - [T] [F] Encoding is computationally more complex than decoding
   - [T] [F] Motion estimation is performed in the DCT domain
   - [T] [F] Temporal redundancy is addressed using DPCM
   - [T] [F] Spatial redundancy is addressed using transform coding

4 Question - 12 pts

Compare the coding efficiency of DCT and KLT when compressing a ECG (electrocardiogram) signal.

- Load the signal `ecg.mat`.
- Segment the signal into non overlapping blocks of B samples ($B = 128$).
- Estimate the KLT transform matrix.
- For each block compute the DCT and the KLT, keeping only the first K samples
- Reconstruct an approximation of the input signal using only the K considered samples
- Plot the SNR for the DCT and the KLT as a function of K (use $K = 0 : 1 : 50$)
- Comment the performance of the DCT vs. KLT. Which transform scheme would you choose for an implementation on a limited-resources wireless sensor? Motivate your answer.